

Mitigating Hallucinations via Epistemic Uncertainty: A Self-Consistency and Entropy Filtering Approach

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<https://github.com/iremcetur/CENG467-Hallucination-Mitigation>

Abstract. Hallucination in large language models (LLMs) remains a critical obstacle for reliable deployment. We investigate whether *epistemic uncertainty*, quantified via Shannon entropy over self-sampled responses, can serve as an effective proxy for answer reliability. We evaluate three baselines on TruthfulQA (817 questions) and HaluEval QA (200-sample subset) using Llama-3.1-8B-Instant via the Groq API. Zero-shot prompting achieves 52.8% accuracy; self-consistency ($N=5$) achieves 53.0%; entropy filtering ($t=0.1$) achieves 62.7% accuracy at 73.2% coverage, a +9.9 percentage-point gain. Entropy bucket analysis confirms the signal: low-entropy questions ($H<0.3$) yield 60.3% accuracy on TruthfulQA and 96.1% on HaluEval, versus 33.3% and 60.0% for high-entropy questions ($H\geq 0.7$). We further validate entropy with AUROC = 0.640 (TruthfulQA) and 0.735 (HaluEval), and abstention F1 = 0.727 and 0.948. Calibration analysis (ECE) shows the confidence signal is well calibrated on HaluEval (0.076) but poorly calibrated on TruthfulQA (0.367); temperature scaling reduces ECE by 82.0% and 69.5% respectively. We additionally test semantic clustering as a refinement, finding it unexpectedly degrades discriminative power on HaluEval (AUROC 0.735 $\rightarrow$ 0.497) because sampled responses are overwhelmingly near-paraphrases regardless of correctness. These results show entropy-based abstention is a simple, training-free mechanism that meaningfully reduces hallucination in black-box LLM settings, while highlighting calibration as a necessary complement to raw entropy filtering.

Keywords: Hallucination mitigation · Epistemic uncertainty · Self-consistency · Shannon entropy · TruthfulQA · HaluEval · AUROC · Model calibration

1 Introduction

Large language models (LLMs) achieve strong performance across NLP tasks but remain prone to hallucination — generating fluent, confident text that is factually incorrect [2]. This poses serious risks in high-stakes domains such as healthcare, law, and education. A key challenge is that LLMs produce point

estimates without any accompanying uncertainty signal: a model is equally fluent whether recalling a fact reliably or confabulating a plausible-sounding falsehood.

This paper investigates whether *epistemic uncertainty* — the model’s internal uncertainty about the correct answer — can be estimated cheaply and used to filter unreliable responses. We adopt the self-consistency framework of Wang et al. [7]: sample $N=5$ independent responses with stochastic decoding and compute the Shannon entropy of the resulting vote distribution as an uncertainty proxy. Our hypothesis is that low-entropy predictions (the model converges to the same answer across samples) are more likely correct than high-entropy predictions (responses scatter). If so, an entropy-based abstention policy should trade coverage for accuracy.

We evaluate this on two complementary benchmarks: TruthfulQA [6] (multiple-choice, targeting common misconceptions) and HaluEval [1] (open-ended QA with paired correct/hallucinated answers), using Llama-3.1-8B-Instant via the Groq API. Beyond the core accuracy-coverage results, we validate entropy with standard uncertainty-quantification metrics (AUROC, F1, ECE) and directly test two proposed refinements — temperature scaling and semantic clustering — rather than leaving them as unexplored future work.

2 Related Work

Ji et al. [2] survey hallucination in NLG, distinguishing intrinsic, extrinsic, and factual hallucination, and identify training data quality and decoding strategy as key contributing factors; our work targets the inference-time question of identifying likely hallucinations without gold references. Wang et al. [7] introduced self-consistency, sampling multiple reasoning paths and aggregating via majority vote, improving accuracy by 10–20% on reasoning benchmarks such as GSM8K; we repurpose the resulting vote-distribution entropy as an uncertainty signal rather than using sampling purely for accuracy gains. Kadavath et al. [3] showed LLMs can verbalize calibrated confidence to a limited extent, though this requires an extra inference pass and can itself hallucinate. Kuhn et al. [5] proposed semantic entropy, clustering semantically equivalent responses before computing entropy to remove surface-form noise; we adopt syntactic majority voting as our primary method (sufficient for multiple-choice and binary correctness) and directly test semantic clustering as an extension in Section 4.4. More broadly, our entropy-filtering approach is an instance of selective prediction [3], where the confidence score derives from sampling variance rather than model logits, making it applicable to black-box API settings.

3 Methodology

3.1 Model, Infrastructure, and Datasets

All experiments use Llama-3.1-8B-Instant via the Groq API, with exponential backoff (45s) on rate-limit errors. **TruthfulQA** [6]: the multiple-choice (mc1)

validation split, 817 questions across 38 misconception categories, average 5.04 choices per question (binary labels, 1=correct); we use mc1 for simplicity of majority-vote aggregation. **HaluEval QA** [1]: 10,000 question-answer pairs with supporting `knowledge`, `question`, `right_answer`, and `hallucinated_answer` fields; we randomly sample 200 examples (seed = 42). Table 1 summarizes both datasets.

Table 1. Dataset comparison summary.

Property	TruthfulQA	HaluEval QA
Total samples	817	10,000 (200 used)
Format	Question + choices + labels	Knowledge + Q + right/halluc.
Avg. choices	5.04 (mc1)	—
Label type	Binary per choice	Implicit (right vs. hallucinated)
Primary use	Accuracy-based evaluation	Pairwise hallucination analysis

3.2 Baselines and Entropy Computation

Baseline 1 (Zero-Shot) queries the model once per question at temperature = 0.0, following the zero-shot chain-of-thought paradigm of Kojima et al. [4], instructing it to respond with a single letter. **Baseline 2 (Self-Consistency, $N=5$)** samples 5 independent completions at temperature = 0.7 and takes the majority vote. Shannon entropy over the vote distribution is:

$$H = - \sum_i p(i) \log p(i), \quad H_{\text{norm}} = \frac{H}{\log K} \quad (1)$$

where $p(i)$ is the vote proportion for option i and K the number of choices; normalizing keeps $H_{\text{norm}} \in [0, 1]$ regardless of K . **Baseline 3 (Entropy Filtering)** answers only when $H_{\text{norm}} < t$, abstaining otherwise; we sweep $t \in \{0.1, \dots, 1.0\}$ and select $t=0.1$, which maximizes accuracy subject to coverage $\geq 50\%$. For **HaluEval**, the model generates free-text answers; correctness uses lexical matching (response must contain $\geq 50\%$ of the reference answer’s content words, length > 3), and entropy is computed over the binary correct/incorrect split across the 5 samples, normalized by $\log 2$.

4 Results

4.1 Baseline Comparison and Entropy Buckets

Table 2 summarizes the three baselines on TruthfulQA. Self-consistency yields only a marginal +0.2% over zero-shot — expected, since TruthfulQA is primarily factual recall rather than multi-step reasoning, limiting the benefit of sample

aggregation. Entropy filtering provides the real gain: answering only the 598 questions (73.2%) where the model is most confident raises accuracy to 62.7% (+9.9 points over zero-shot).

Table 2. Baseline comparison on TruthfulQA (817 questions).

Method	Correct	Total	Accuracy	Coverage
Baseline 1 — Zero-Shot ($t=0.0$)	431	817	52.8%	100%
Baseline 2 — Self-Consistency ($N=5$)	433	817	53.0%	100%
Baseline 3 — Entropy Filtering ($t=0.1$)	375	598	62.7%	73.2%

Table 3 confirms the entropy signal directly: low-entropy TruthfulQA questions reach 60.3% accuracy versus 26.3% for medium entropy (the $n=12$ high-entropy bucket is too small to be reliable). On HaluEval the pattern is sharper still (96.1% vs. 60.0%), and 90% of samples fall in the low-entropy bucket — knowledge-grounded questions tend to have clear, consistently-reproduced answers.

Table 3. Accuracy by normalized entropy bucket (H_{norm}).

Entropy Bucket	TQA n	TQA Acc.	Halu n	Halu Acc.
Low ($H_{\text{norm}} < 0.3$)	638	60.3%	180	96.1%
Medium ($0.3 \leq H_{\text{norm}} < 0.7$)	167	26.3%	—	—
High ($H_{\text{norm}} \geq 0.7$)	12	33.3%	20	60.0%

Figure 1 shows the accuracy–coverage trade-off as t varies; stricter thresholds raise accuracy but reduce coverage, degrading to the Baseline-2 level (53.0%) once all questions are answered. Figure 2 shows the entropy-bucket pattern holds across both task formats.

4.2 Additional Evaluation Metrics

Accuracy and coverage alone do not establish that entropy is a *statistically reliable* predictor or that it is *calibrated*. We report AUROC (discriminative power), Precision/Recall/F1 (the abstention decision as binary classification), and Expected Calibration Error (ECE).

AUROC. Using $-H_{\text{norm}}$ as a score, we measure how well it separates correct from incorrect responses (0.5 = random, 1.0 = perfect). Table 4 shows both datasets clear the random baseline, with HaluEval’s cleaner (near-binary) entropy distribution yielding stronger separation. Figure 3 shows the full ROC curves.

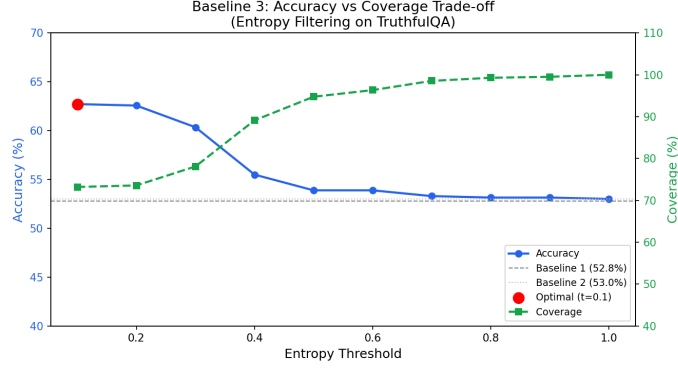


Fig. 1. Accuracy vs. coverage trade-off for entropy filtering on TruthfulQA. Red dot: optimal point ($t=0.1$). Dashed lines: Baseline 1/2.

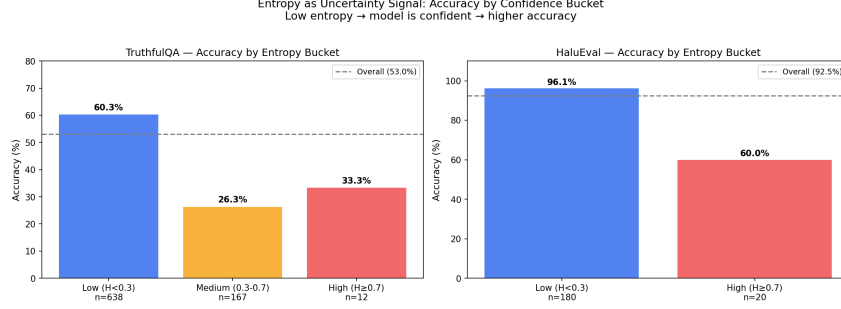


Fig. 2. Accuracy by entropy bucket, TruthfulQA (left) vs. HaluEval (right).

Precision/Recall/F1. Framing entropy filtering at $t=0.1$ as a binary classifier (Table 5): on TruthfulQA, recall (0.866) exceeds precision (0.627) — the filter answers most correct questions but also lets through many confidently-wrong ones (Section 4.3). On HaluEval both exceed 0.93, reflecting the cleaner signal there.

ECE. We assess whether $1 - H_{\text{norm}}$ is a well-calibrated confidence score using 10 equal-width bins: $\text{ECE} = \sum_b \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|$. HaluEval is reasonably well calibrated (0.076); TruthfulQA is not (0.367) — confidence systematically overstates accuracy (Table 6, Figure 4). This means entropy *discriminates* effectively (Table 4) but is not directly interpretable as a calibrated probability on TruthfulQA.

4.3 Error Analysis

Among the 598 TruthfulQA questions answered at $t=0.1$, 375 are correct and 223 (37.3%) are **false-confidence** errors: the model converges on a wrong answer

Table 4. AUROC of normalized entropy as a correctness predictor.

Dataset	N	N Correct	AUROC
TruthfulQA	817	433	0.640
HaluEval	200	185	0.735

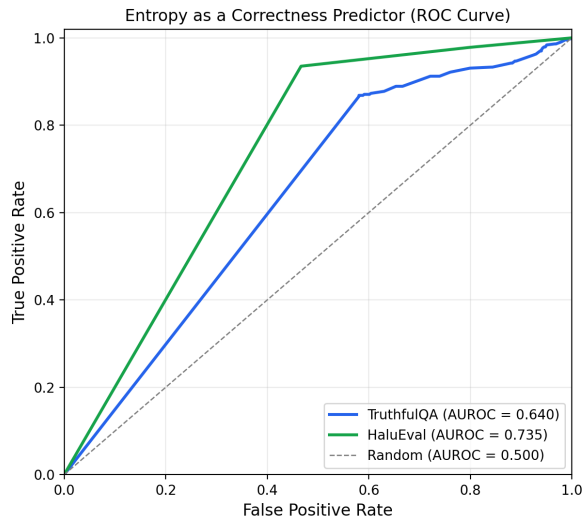


Fig. 3. ROC curves: entropy discriminating correct/incorrect responses.

with full agreement across samples (Table 7). Manual inspection shows these consistently reflect intuitive but factually wrong options — e.g., misattributing “I cannot tell a lie” or misidentifying the world’s smallest country — i.e., misconceptions prevalent in the pretraining corpus itself, where low sampling variance reflects consistency, not correctness. The complementary cost is **high-entropy correct** cases: questions the model would have answered correctly but where votes split enough to trigger abstention, visible in the gap between Baseline 2’s 53.0% overall accuracy and the 86.6% recall in Table 5.

This 37.3% rate quantifies why calibration matters: entropy discriminates well in aggregate (AUROC = 0.640) but full sample agreement gives no further signal to flag these residual errors — addressing them likely requires orthogonal techniques such as retrieval augmentation or external fact verification.

4.4 Extended Analyses

Rather than leaving calibration and semantic clustering as unexplored future work, we test both directly using the existing sampled responses (no new API calls).

Table 5. Precision, Recall, F1 of entropy-based abstention at $t=0.1$.

Dataset	Precision	Recall	F1
TruthfulQA	0.627	0.866	0.727
HaluEval	0.961	0.935	0.948

Table 6. Expected Calibration Error (ECE) of $1 - H_{\text{norm}}$.

Dataset	ECE
TruthfulQA	0.367
HaluEval	0.076

Temperature scaling. We fit a single scalar T minimizing negative log-likelihood on the correctness labels, rescaling confidence as $\sigma(\text{logit}(c)/T)$. This reduces ECE by 82.0% on TruthfulQA and 69.5% on HaluEval (Table 8, Figure 5). The large fitted T on TruthfulQA (27.23) indicates the raw confidence needed substantial flattening toward the base accuracy rate — consistent with its weaker AUROC relative to HaluEval.

Semantic clustering. Kuhn et al. [5] argue semantically equivalent responses with different wording should count as agreement. We embed each HaluEval sample’s 5 responses with `all-MiniLM-L6-v2` and apply agglomerative clustering (cosine distance, threshold = 0.3) before recomputing entropy over the cluster distribution. The result is counter to expectation: semantic entropy *degrades* discrimination to near-random (Table 9). Inspection shows the mean clusters per sample was only 1.07/5 — the model’s sampled responses are nearly always semantically equivalent paraphrases, even when lexical matching marks the majority answer wrong. The model often expresses the *same* (sometimes incorrect) claim in different words, so clustering collapses nearly every sample into one cluster regardless of correctness, destroying the variance the syntactic signal relies on. For this task, syntactic disagreement is itself informative about genuine content uncertainty, and semantic clustering does not straightforwardly transfer the benefits reported by Kuhn et al.

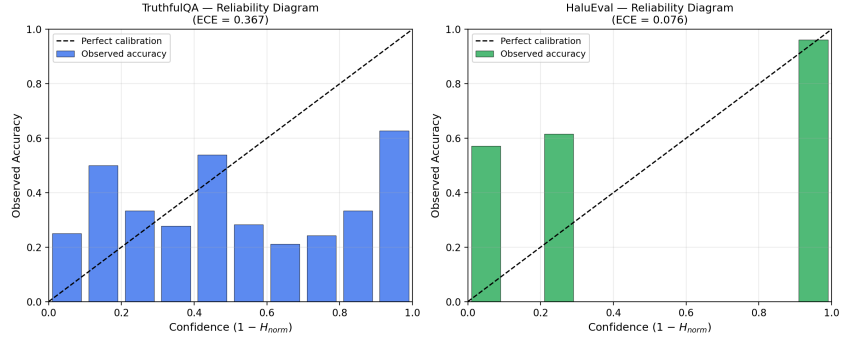


Fig. 4. Reliability diagrams: TruthfulQA (ECE = 0.367) vs. HaluEval (ECE = 0.076).

Table 7. Outcome distribution on TruthfulQA at $t=0.1$ (598 answered).

Outcome	Count	% of answered
Correct	375	62.7%
False confidence (wrong, low H)	223	37.3%

Table 8. Effect of temperature scaling on ECE.

Dataset	Fitted T	ECE before	ECE after
TruthfulQA	27.23	0.367	0.066
HaluEval	4.43	0.076	0.023

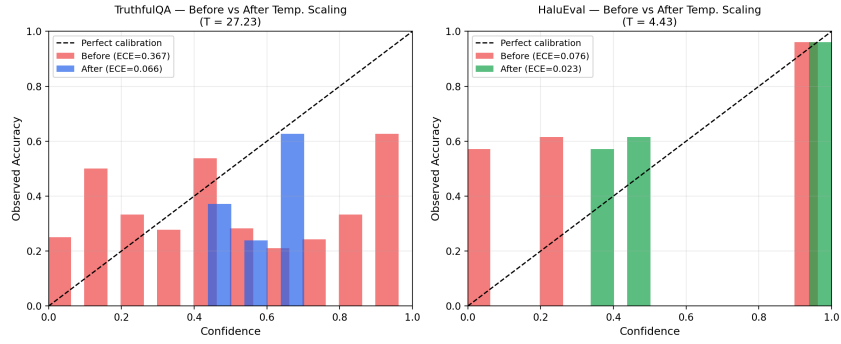


Fig. 5. Reliability diagrams before/after temperature scaling.

Table 9. Syntactic vs. semantic entropy as a correctness predictor (HaluEval).

Entropy variant	AUROC
Syntactic (binary correct/incorrect)	0.735
Semantic (embedding clusters)	0.497

5 Discussion

Normalized Shannon entropy over self-sampled responses is a practical hallucination proxy: when a model has reliably encoded a fact, its outputs are robust to decoding perturbations (low entropy); when it must interpolate between imperfectly learned patterns, sampling variance — and entropy — rise.

Limitations. (1) $N=5$ is a small ensemble relative to the $N=20\text{--}40$ typical in prior work; this was constrained by Groq API rate limits and would likely sharpen entropy estimates if increased. (2) The HaluEval correctness criterion (lexical matching) is a noisy proxy for semantic correctness; embedding-based similarity or an LLM judge would be more robust. (3) 26.8% of TruthfulQA questions are abstained at $t=0.1$, requiring a fallback strategy in real deployment. (4) We directly tested semantic clustering (Section 4.4) and found it *underperforms* our syntactic approach on HaluEval (AUROC 0.735 $\rightarrow$ 0.497), since responses are overwhelmingly near-paraphrases; this finding may not generalize to settings with more lexically diverse outputs. (5) Raw entropy-derived confidence is poorly calibrated on TruthfulQA (ECE=0.367); we addressed this with temperature scaling (Section 4.4), cutting ECE by 82.0%/69.5%, and recommend such post-hoc correction for applications requiring calibrated probabilities rather than raw entropy scores.

6 Conclusion

Normalized Shannon entropy over self-sampled responses is an effective, training-free proxy for epistemic uncertainty in LLMs. Entropy-based abstention on TruthfulQA yields a +9.9 percentage-point accuracy gain over zero-shot while answering 73.2% of questions; on HaluEval the signal is stronger still. Beyond accuracy-coverage, we validated entropy with AUROC (0.640/0.735) and F1 (0.727/0.948), and directly tested two proposed refinements rather than leaving them speculative: temperature scaling substantially improved calibration (ECE $-82.0\%/ -69.5\%$), while semantic clustering unexpectedly degraded discrimination on HaluEval, since sampled responses are overwhelmingly near-paraphrases regardless of correctness.

These results position entropy filtering as a practical first-line defense against hallucination in black-box LLM deployments, but not a complete solution: low entropy indicates the model is *consistent*, not that it is *correct*, particularly on benchmarks like TruthfulQA that target ingrained misconceptions — illustrated by the 37.3% false-confidence rate among answered questions. Remaining future work includes larger sample sizes ($N\geq 20$, beyond our API budget here), combination with retrieval-augmented generation or external fact verification to catch high-confidence hallucinations that sampling-based uncertainty cannot detect, and extension to generation tasks beyond factual QA.

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